# Transparency in Debt Crises

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# Motivation

*"The international community learned lessons from the Mexican crisis too. The most important was the* ***need for transparency****, especially as regards economic data. The IMF and investors simply did not know what was happening to Mexico's reserves in the lead-up to the crisis."* - Stanley Fischer

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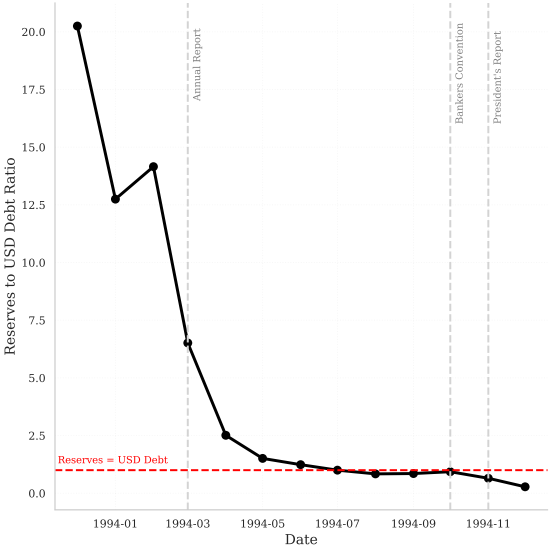
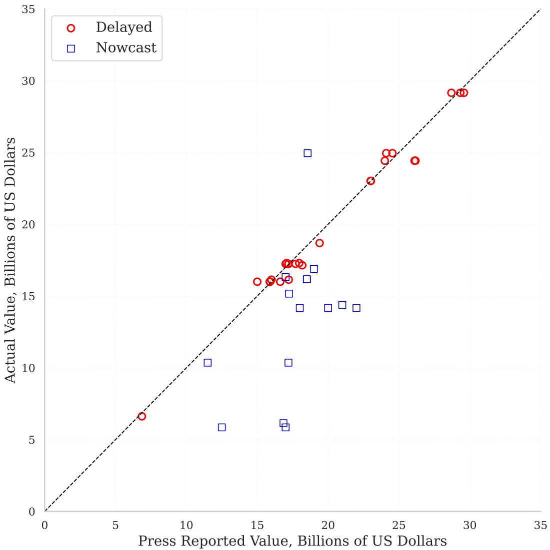
**Disclosure policies in emerging market economies:**

- Greek misreports before the European Debt Crisis.
- Hidden debt and swap lines with China.
- Bolivian reserve reports (2024).
- **This paper:** Mexico Peso Crisis 1994-1995

# Dollar Reserves vs. Dollar Debt



# Reports vs. Actual Reserve Levels



# This Paper

Literature

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**The trade-off:**

- By fully disclosing:
  - Lower spreads
  - Issuance according to repayment capacity
- By not disclosing:
  - more debt at a low spread in bad times.
  - less debt at a high spread in good times

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**Results:**

- Eaton-Gersovitz timing + endogenous information structure.
- $\downarrow$ deadweight loss from default, $\uparrow$ incentives not to disclose information ex-ante.
- **Takeaway:** Non-disclosure functions as insurance for bad times.

# Model

# Setup

- Three periods: $t = 0, 1, 2$
- Continuum of competitive risk-neutral lenders with discount rate $1/R$
- Sovereign
  - Commits to a disclosure policy $\phi \in \{0, 1\}$ at $t = 0$
  - Has a constant endowment of $y$, and zero debt due at $t = 1$
  - Cannot commit to repay at $t = 2$
  - Discounts the future at a rate $\beta < 1/R$
- Endowment after repayment at $t = 2$ can be $i = H, L$ with $\theta_H > \theta_L$:

$$\theta_i = \begin{cases} \theta_H, & \text{with prob. } 1 - \pi \\ \theta_L, & \text{with prob. } \pi \end{cases}$$

- A sovereign that defaults gets $y_d \leqslant \theta_L$

# Timing

| $t = 0$ | $t = 1$ | $t = 2$ |
|---|---|---|
| 1) Choice of disclosure policy $\phi \in \{0, 1\}$ | 1) Government learns its type $(\theta_i)$<br>2) Govt type is revealed $(\phi = 1)$ or not $(\phi = 0)$<br>3) Choice of debt<br>4) Price is determined $q$<br>5) Consumption takes place | 1) Default decision<br>2) Consumption |

# Problem under complete information

**Budget constraints:**

$$c_1(\theta) = y + q(\theta, b(\theta))b(\theta), \quad \forall\theta \qquad (t = 1)$$

$$c_2(b,\theta) = \delta(b,\theta) \times y_d + \Big(1 - \delta(b,\theta)\Big) \times (\theta - b), \quad \forall(b,\theta) \qquad (t = 2)$$

**Type-$\theta$ sovereign's problem:**

$$V_{FD}(\theta; y_d) = \max_{b(\theta)} \left\{ \ln\Big(y + q(b,\theta)b(\theta)\Big) + \beta \ln\Big(c_2^*(b,\theta)\Big) \right\}$$

subject to:

$$c_2^*(b,\theta) = \max_{\delta\in\{0,1\}} \delta \times y_d + (1-\delta) \times (\theta - b)$$

**Investor's profits:**

$$\frac{1}{R}\left[\delta(\theta,b) \times 0 + (1 - \delta(\theta,b))\right] - q\,(\theta,b) = 0, \quad \forall(b,\theta)$$

# Complete Information Benchmark

**Equilibrium**

An **equilibrium under complete information** is a strategy for the sovereign $(\delta_{FD}(b, \theta), b_{FD}(\theta))$ and a price schedule for the lenders $q_{FD}(b, \theta)$ such that:

i) The default policy $\delta_{FD}(\theta, b)$ maximizes consumption at $t = 2$.
ii) Given $q_{FD}(b, \theta)$, the choice of debt $b_{FD}(\theta)$ maximizes the sovereign's lifetime payoff.
iii) Investors make zero profits.

# Characterization

**Lemma**

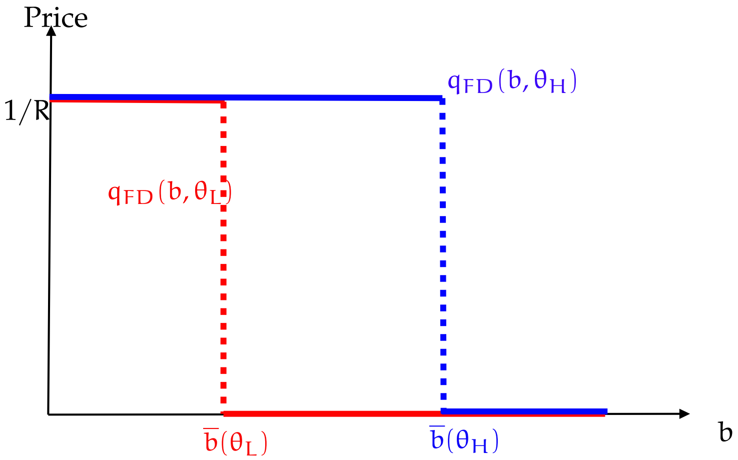
Define $\overline{b}(\theta) \equiv \theta - y_d$. The following strategy and price schedule is an **equilibrium** under full information:

$$\delta_{FD}(b,\theta) = \begin{cases} 1, & \text{if } b > \overline{b}(\theta) \\ 0, & \text{if } b \leqslant \overline{b}(\theta) \end{cases}$$

$$q_{FD}(b,\theta) = \begin{cases} 0, & \text{if } b > \overline{b}(\theta) \\ 1/R, & \text{if } \mathbf{b} \leqslant \overline{\mathbf{b}}(\theta) \end{cases}$$

$$\mathbf{b}_{FD}(\theta) = \min\left\{ \frac{\theta - \beta R y}{1+\beta}, \overline{\mathbf{b}}(\theta) \right\}$$

# Equilibrium Prices

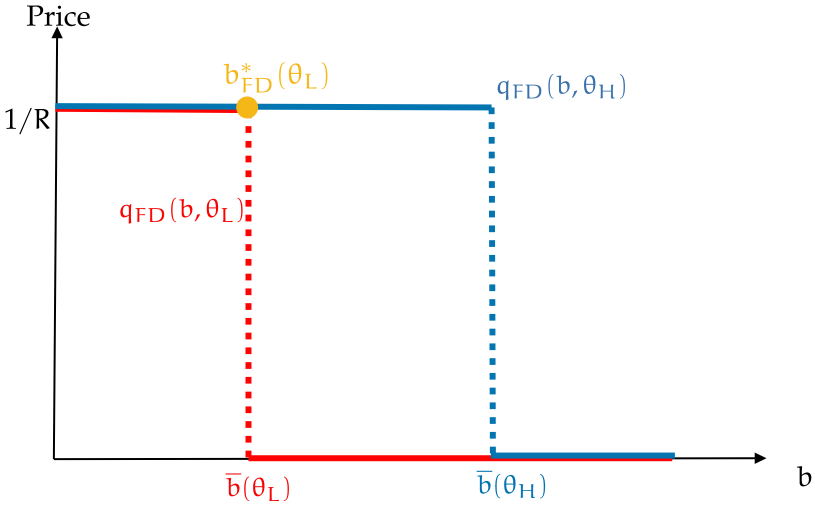


# Parametric Assumption

**Assumption 1**

$$y_d > \frac{\beta R}{1+\beta}\left(y + \frac{\theta_L}{R}\right)$$

# L-type's choice of debt under Assumption 1



# Problem under incomplete information

**Budget constraints:**

$$
\begin{aligned}
c_1(\theta) &= y + q(b(\theta))b(\theta), \quad \forall\theta && (t=1)\\
c_2(b,\theta) &= \delta(b,\theta) \times y_d + \Big(1 - \delta(b,\theta)\Big) \times (\theta - b), \quad \forall(b,\theta) && (t=2)
\end{aligned}
$$

**Type-$\theta$ sovereign's problem:**

$$
V_{ND}(\theta; y_d) = \max_{b(\theta)} \left\{ \ln\Big(y + q\big(b(\theta)\big)b(\theta)\Big) + \beta \ln\Big(c_2^*(b,\theta)\Big) \right\}
$$

subject to:

$$
c_2^*(b,\theta) = \max_{\delta \in \{0,1\}} \delta \times y_d + (1-\delta) \times (\theta - b)
$$

**Investor's expected profits:**

$$
\frac{1}{R}\mathbb{E}_\mu\left[\delta(b) \times 0 + (1 - \delta(b))\right] - q(b(\theta)) = 0, \quad \forall b
$$

where $\boldsymbol{\mu(b) \equiv P(\theta = \theta_L | b)}$

# Incomplete Information Problem

- Pure strategies (for now)

**Definition**

A **Perfect Bayesian Equilibrium (PBE)** is a strategy for the sovereign $(b_{ND}(\theta), \delta_{ND}(b, \theta))$ and a price schedule for the lender $q_{ND}(b)$, and beliefs $\mu(b)$ such that:

1) The default policy $\delta_{ND}(b, \theta)$ maximizes the sovereign's consumption at $t = 2$.
2) Given $q_{ND}(b)$ the choice of debt maximizes sovereign's lifetime payoffs.
3) Beliefs follow the Bayesian update whenever possible.
4) Investors make zero profits.

# Separating Equilibrium

**Definition**

A **separating equilibrium** is a PBE in which both types pick different levels of debt: $\mathbf{b}_{\mathbf{ND}}(\theta_L) \neq \mathbf{b}_{\mathbf{ND}}(\theta_H)$, so that $\mu(b_{ND}(\theta_L)) = 1$ and $\mu(b_{ND}(\theta_H)) = 0$.

# Separating Equilibrium

**Proposition** Proof

Under **Assumption 1**, there is no separating equilibrium.

# Pooling Equilibra

**Definition**

A **pooling equilibrium** is a PBE in which both types pick the same debt level:

$$b_{ND}(\theta_L) = b_{ND}(\theta_H) = \tilde{b}$$

Let $\tilde{b}$ be:

$$\tilde{b} \equiv \min \left\{ \frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}, \theta_H - y_d \right\}$$

Consider the following belief system and default policy:

$$\mu(b) = \begin{cases} \pi, & \text{if } b \leqslant \tilde{b} \\ 1, & \text{if } b > \tilde{b} \end{cases}$$

$$q_{ND}(b) = \begin{cases} 1/R, & \text{if } b < \theta_L - y_d \\ \frac{1-\mu(b)}{R}, & \text{if } b \in (\theta_L - y_d, \theta_H - y_d] \\ 0, & \text{if } b > \theta_H - y_d \end{cases}$$

$$\delta_{ND}(b, \theta) = \begin{cases} 1, & \text{if } b > \overline{b}(\theta) \\ 0, & \text{if } b \leqslant \overline{b}(\theta) \end{cases}$$

# Risky vs. Safe pooling

Two candidates of pooling equilibrium:

- *Risky pooling:* both types pick $\tilde{b}$, the L-type defaults.
- *Safe pooling:* both types pick $\theta_L - y_d$, both types repay.

Define the interval of $y_d$ in which only the L-type is constrained:

$$y_d \in \left( \frac{\beta R}{1+\beta}\left(y + \frac{\theta_L}{R}\right), \min\left\{\theta_L, \frac{\beta R}{1+\beta}\left(y + \frac{\theta_H}{R}\right)\right\}\right) \equiv Y_d$$

**Proposition** Proof

Let $y_d \in Y_d$. There exists $\pi^*$ such that for $\pi < \pi^*$, the unique H-type preferred PBE is the risky pooling equilibrium. Moreover,

$$\frac{\partial \pi^*}{\partial y_d} > 0$$

# Disclosure choice

Consider the following condition

$$\log\left(\frac{Ry+\theta_H}{Ry+\theta_L}\right)+\frac{R\beta y-\theta_H}{Ry+\theta_H}>0 \qquad (\star)$$

Define:

$$\Delta(\pi)\equiv V_{ND}(\pi)-V_{FD}(\pi)$$

**Proposition**

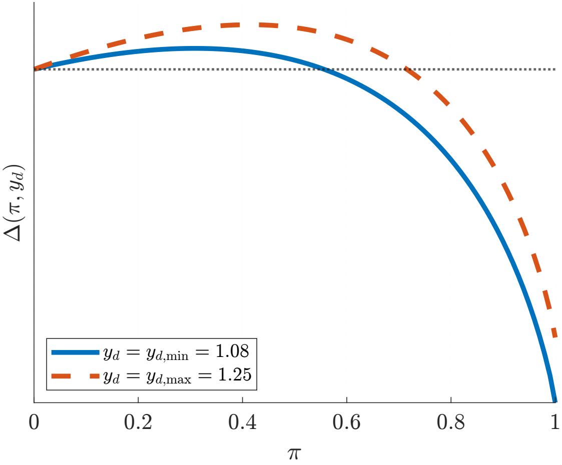
Suppose $(\star)$, holds. Then, for any $y_d \in Y_d$, there exists $\epsilon(y_d)>0$ such that

$$\Delta(\pi)>0, \quad \text{for all } \pi\in(0,\epsilon(y_d))$$

That is, non-disclosure (ND) is preferred over full disclosure (FD).

# Disclosure choice

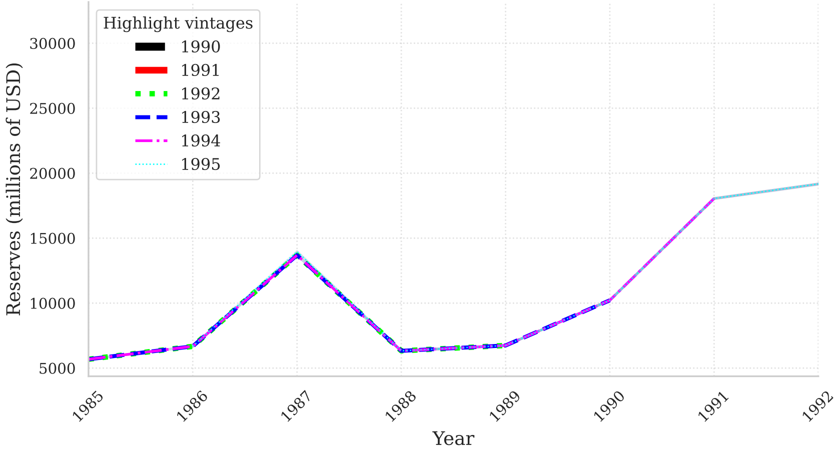
- L-type gains from pooling **(insurance)**
- H-type loses from pooling (adverse selection discount)



# Mexican Peso Crisis (1994-1995)

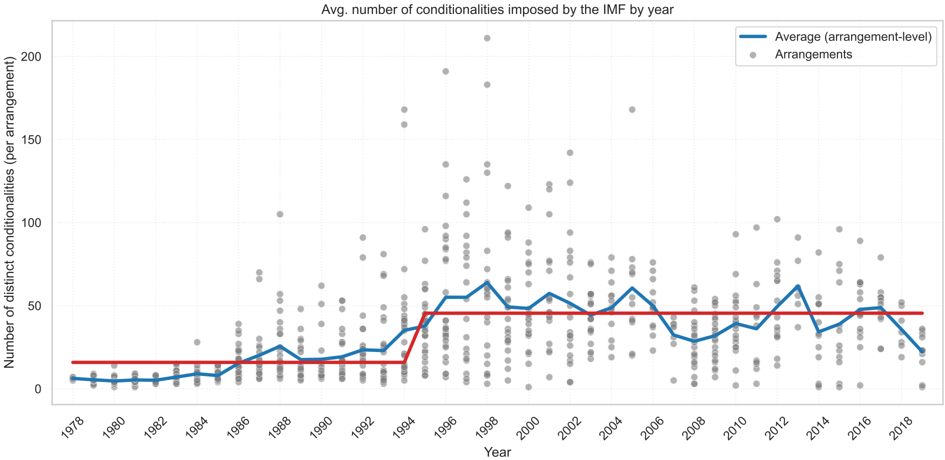
# Truth-telling

Similar reserve reports across years for different vintages.
Consistent with truthful disclosure.



# IMF conditionality

Increase in the number of conditionalities in IMF deals
Consistent with an increase in default costs



$$\texttt{Number of conditions}_t = \beta_0 + \beta_1 \texttt{Post-1995}_t + \epsilon_t,$$

Table: OLS: Post-1995 shift in conditionalities

| **Coefficient** | **Value** |
|---|---|
| $\beta_0$ | 15.59*** |
| | (1.30) |
| $\beta_1$ | 30.02*** |
| | (1.80) |
| $R^2$ | 0.26 |
| Observations | 793 |

Std. errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

# Conclusions

- Governments often use informational advantages against investors
- To capture its incentives we built a default model with an endogenous choice of information structure
- No-disclosure creates an insurance opportunity, preferred when deadweight losses are low enough
- Increase in transparency for the Mexican case is consistent with an increase in IMF conditionalities

# Appendix

# Literature

Back

1. **Sovereign debt under private information**

   Sandleris (2008), Zabai (2014), Perez (2017), Szkup (2021), Amador and Phelan (2021), Guler et. al (2021), Horn et. al (2024)

2. **Role of information in financial crises**

   Gorton and Ordonez (2020), Faria-e-Castro et. al (2017),

3. **Transparency in coorporations**

   Jehiel (2015), Kamenica and Gentzkow (2011), Dye (1983), Verrecchia (1983)

# Proof for Proposition 1 

Back

**Proof by contradiction:**

- Let $\left(\mathbf{b}_{\mathbf{ND}}(\theta), \delta_{\mathbf{ND}}(\mathbf{b},\theta), \mathbf{q}_{\mathbf{ND}}(\mathbf{b}), \boldsymbol{\mu}(\mathbf{b})\right)$ be a separating equilibrium.
- By definition $\mathbf{b}_{\mathbf{ND}}(\theta_{\mathbf{L}}) \neq \mathbf{b}_{\mathbf{ND}}(\theta_{\mathbf{H}})$, with $\mu\left(b_{ND}(\theta_L)\right) = 1$ and $\mu\left(b_{ND}(\theta_H)\right) = 0$.
- Since investors know with which type they are dealing with prices are $q(b) \in \{0, 1/R\}$ for $b \in \{b_{ND}(\theta_L), b_{ND}(\theta_H)\}$, due to the zero profit condition.
- Note that regardless of $\mu$, $(\theta_L - y_d, 1/R)$ is always available for both types.
- Therefore, it must be the case that $q(b) = 1/R$ for $b \in \{b_{ND}(\theta_L), b_{ND}(\theta_H)\}$.
- It then follows from **Assumption 1** that $b_{ND}(\theta_L) = \theta_L - y_d$.
- Since $b_{ND}(\theta_H) < \theta_L - y_d$ is not optimal for the H-type, and $q(b_{ND}(\theta_H)) = 1/R$, it must be that $\theta_H - y_d \geqslant b_{ND}(\theta_H) > \theta_L - y_d$.
- But this is a profitable deviation for the L-type, that would prefer to mimic, therefore $\left(\mathbf{b}_{\mathbf{ND}}(\theta), \delta_{\mathbf{ND}}(\mathbf{b},\theta), \mathbf{q}_{\mathbf{ND}}(\mathbf{b}), \boldsymbol{\mu}(\mathbf{b})\right)$ cannot be a separating equilibrium.

# A Mechanism Design Problem (1/2)

Back

**Best Default Outcome**

$$\max_{\{\tilde{b},\tilde{q}\}} u(y+\tilde{q}\tilde{b}) + \beta u(\theta_H - \tilde{b}) \tag{P}$$

subject to

$$\tilde{b} \leqslant \theta_H - y_d \qquad (\text{SUST-H}, \nu)$$

$$\tilde{b} \geqslant \theta_L - y_d \qquad (\text{ICD-L}, \chi)$$

$$u(y+\tilde{q}\tilde{b}) + \beta u(y_d) \geqslant u(y) + \beta u(\theta_L) \qquad (\text{PC-L}, \eta)$$

$$\underbrace{\pi \times \left[-\tilde{q}\tilde{b}\right] + (1-\pi) \times \left[\frac{\tilde{b}}{R} - \tilde{q}\tilde{b}\right]}_{\frac{(1-\pi)}{R}\tilde{b} - \tilde{q}\tilde{b}} \geqslant 0 \qquad (\text{PC-I}, \xi)$$

# Efficient Default Allocation

Back

**Lemma**

Suppose $y_d \leqslant \theta_L$, and the L-type defaults, that is, $\tilde{b} > \theta_L - y_d$ ($\chi = 0$). Therefore, investors make zero profits ($\xi > 0$), that is,

$$\tilde{q} = \frac{1-\pi}{R}$$

*Proof.* Let $(\tilde{q}, \tilde{b})$ be a solution to (P). Assume, by contradiction, that investors make positive profits, that is, $\tilde{q} < \frac{1-\pi}{R}$. Since the H-type's payoff is strictly increasing in $\tilde{q}$, it must be the case that there exists $q' > \tilde{q}$ that increases the H-type's payoff. Since the payoff of type L is also increasing in the price of the debt, and when $\tilde{q}$ PC-L is satisfied, we must have that $q'$ also satisfies PC-L, which means that $\tilde{q}$ is not a solution to (P).

**Proposition** Back

The risky pooling contract maximizes the payoff of the H-type among all contracts in which the L-type defaults.

*Proof.*

$$(1+\eta)u'\left(y+\frac{1-\pi}{R}\times\tilde{b}\right)(1-\pi)-R\beta u'(\theta_H-\tilde{b})=R(\nu-\chi) \tag{1}$$

By looking for a solution in which the L-type, defaults, that is, $\tilde{b}\in(\theta_L-y_d,\theta_H-y_d)$, we have that $\nu=\chi=0$. If the L-type's PC-I binds but PC-L is slack ($\eta=0$), we have:

$$\tilde{b}=\frac{\theta_H-\frac{R\beta}{1-\pi}\times y}{1+\beta}$$

We now look for a parametrization in which $\eta=0$. Notice that if $y_d=\theta_L$, PC-L is slack. By continuity of $u\left(y+\frac{1-\pi}{R}\tilde{b}\right)+\beta u(y_d)$ in $y_d$, there must exist $\epsilon>0$ such that PC-L holds with strict inequality for $y_d\in(\theta_L-\epsilon,\theta_L)$. Define:

$$\overline{\epsilon}\equiv\sup\{\epsilon:\epsilon>0,\text{ such that PC-L holds with strict inequality for } y_d=\theta_L-\epsilon\}$$

By taking $y_d\equiv\theta_L-\overline{\epsilon}$ we have that $\eta=0$.

# A Mechanism Design Problem (2/2)

Back

**Best Repayment Outcome**

$$\max_{\{b_H, b_L, q_H, q_L\}} u(y + q_H b_H) + \beta u(\theta_H - b_H) \tag{S}$$

subject to

$$b_H \leqslant \theta_H - y_d \qquad (\text{SUST-H}, \nu_H)$$

$$b_L \leqslant \theta_L - y_d \qquad (\text{SUST-L}, \nu_L)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geqslant u(y + q_H b_H) + \beta u(y_d) \qquad (\text{ICD-L}, \chi)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geqslant u(y + q_H b_H) + \beta u(\theta_L - b_H) \qquad (\text{ICR-L}, \lambda)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geqslant v_L^a \qquad (\text{PC-L}, \eta)$$

$$b_L \times \left[\frac{1}{R} - q_L\right] \geqslant 0 \qquad (\text{PCI-L}, \xi_L)$$

$$b_H \times \left[\frac{1}{R} - q_H\right] \geqslant 0 \qquad (\text{PCI-H}, \xi_H)$$

Back

**Proposition**

The safe pooling outcome maximizes the H-type's payoff in which: (i) lenders make zero profits; (ii) the sustainability constraint for the L-type binds; (iii) the L-type prefers to default rather than repay the H-type's contract.

*Proof.*

We look for a solution with $\xi_H, \xi_L > 0$, which implies $q_H = q_L = 1/R$. By the FOC with respect to $(b_H)$:

$$u'\left(y + \frac{b_L}{R}\right) - R\beta u'(\theta_L - b_L) = \frac{R\nu_L}{\chi + \lambda + \eta}$$

Consider first the case in which the sustainability constraint binds for the L-type: $\nu_L > 0$, that is, $b_L = \theta_L - y_d$. We can then rewrite the ICD-L as:

$$u\left(y + \frac{\theta_L - y_d}{R}\right) \geqslant u\left(y + \frac{b_H}{R}\right) \implies b_H \leqslant \theta_L - y_d$$

Since SUST-L binds we know that the L-type is indifferent between default and repayment of $b_L$. By the FOC with respect to $b_H$:

$$u'(y + b_H/R) - R\beta u'(\theta_H - b_H) = \chi u'(y + b_H/R) + \lambda[u'(y + b_H/R) - R\beta u'(\theta_L - b_H)]$$

If the relevant constraint is ICD-L (picking the H-type's contract and defaulting is better than picking the H-type's contract and repaying) it must be the case that:

$$u(y_d) \geqslant u(\theta_L - b_H)$$

That is:

$$b_H \geqslant \theta_L - y_d$$

Back

# Pooling Eq. - General Case

Back

**Only L-type is constrained:**

$$y_d \in \left( \frac{\beta R}{1+\beta}\left(y + \frac{\theta_L}{R}\right), \frac{\beta R}{1+\beta}\left(y + \frac{\theta_H}{R}\right)\right)$$

**Lemma**

There exists $\pi_L(y_d)$ such that, for $\pi < \pi_L(y_d)$, the L-type prefers the *risky pooling* debt level

*Proof.*

$$y + \frac{1-\pi}{R}\left[\frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}\right] > y + \frac{\theta_L - y_d}{R}$$

$$\underbrace{1 - \left[\frac{(1+\beta)(\theta_L - y_d) + \beta R y}{\theta_H}\right]}_{\equiv \pi_L(y_d)} > \pi$$

# Pooling Eq. - General Case

Back

Define the risky pooling payoff for the H-type:

$$\mathcal{F}(\pi) \equiv (1+\beta)\log\left(\frac{y + \frac{1-\pi}{R}\times\theta_H}{1+\beta}\right) + \beta\log\frac{\beta R}{1-\pi}$$

**Lemma**

$\mathcal{F}$ is strictly convex. Moreover, at its minimum $\rho'$:

$$\mathcal{F}(\rho') = \log(y) + \beta\log(\theta_H)$$

# Pooling Eq. - General Case

Back

**Lemma**

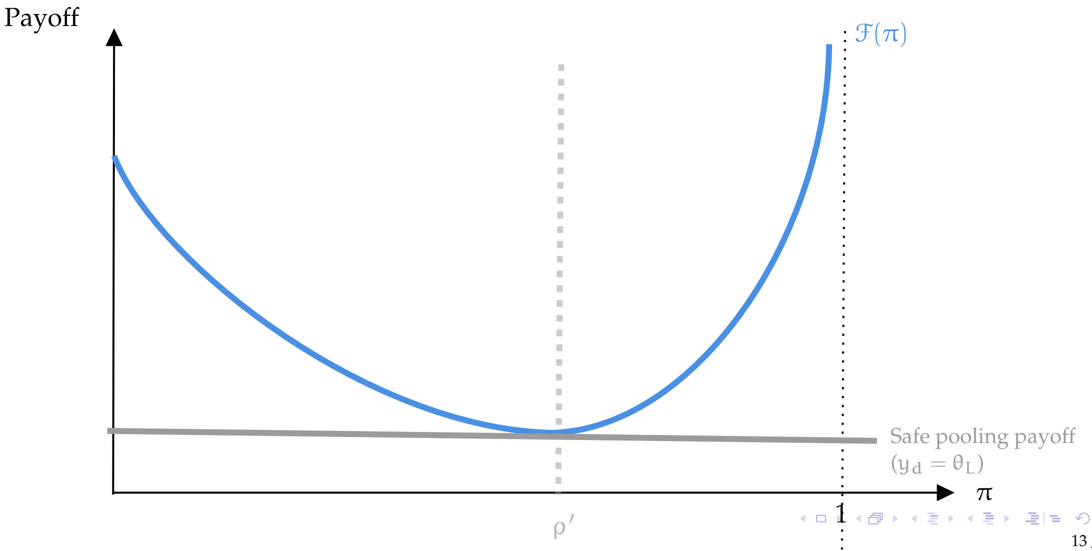
There exists $\pi_H(y_d)$ such that, for $\pi < \pi_H(y_d)$, the H-type prefers the *risky pooling* debt level

**Proposition**

Suppose only the L-type is constrained. There exists $\pi^*$ such that for $\pi < \pi^*$, the unique pooling equilibrium has $\tilde{b} = \frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}$

# Pooling Eq. - General Case

Back



# Pooling Eq. - General Case

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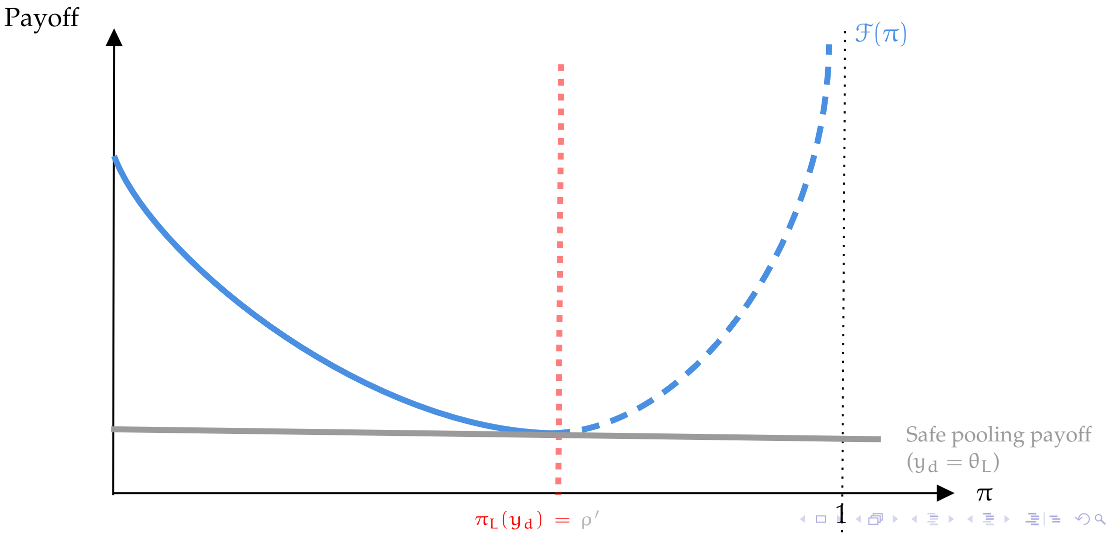

# Pooling Eq. - General Case

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